# Week 2: Data Warehouse Overview

Keeyong Han

# Table Of Contents


1. Recap of 1st Week
2. Definition of Data Warehouse
3. OLTP vs. OLAP
4. ETL vs. ELT
5. Data Warehouse vs. Data Lake
6. Break
7. Data Warehouse Options
8. Sneak Peek at Snowflake
9. Example Data System
10. Lab1/Demo/Homework

# Recap of 1st Week

# Data Flow

Site traffic &
3rd party data

1

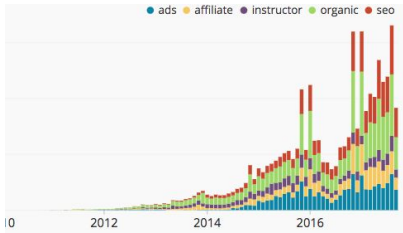
ELT

ETL

Data warehouse

Data Team

2




Business
Insights

3

Site Services

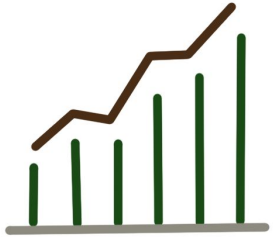
Scalable Product Improvement

# Data System Evolution

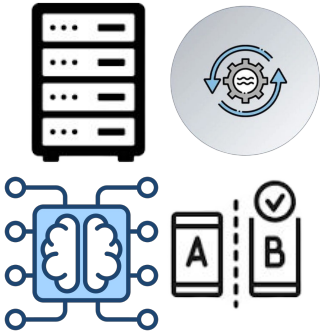
Data Sources

Data Pipelines

Data Storages



BI Tools



# Study Materials

- You can read these from oreilly.com for free:

1. AI Engineering: Building Applications with Foundation Models, Chip Huyen
2. Designing Machine-Learning Systems, Chip Huyen
3. Designing Data-Intensive Applications 2nd edition: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems, Martin Kleppmann.
4. The Data Warehouse Toolkit, 3rd Edition: The Definitive Guide to Dimensional Modeling, Ralph Kimball and Margy Ross.

- SQL: https://leetcode.com/studyplan/top-sql-50/
- Python: https://www.udemy.com/course/100-days-of-code/

# Office Hours & Classroom & Snowflake Academia

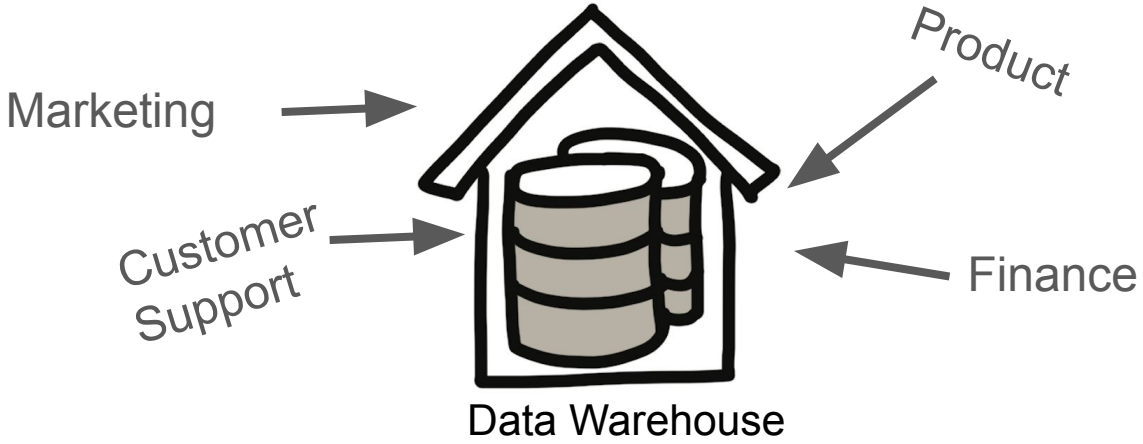
- Instructor: Keeyong Han (keeyong.han@sjsu.edu):
  - 2 to 3PM Thursday at ISB 863
  - 2 to 3PM Friday at ISB 863
- TA: Jeff Chong
  - 3 to 4PM Fridays

- For Snowflake Academia, make sure to store your password :)
- Also if you haven’t got any email about the account, let me know

# Definition of Data Warehouse

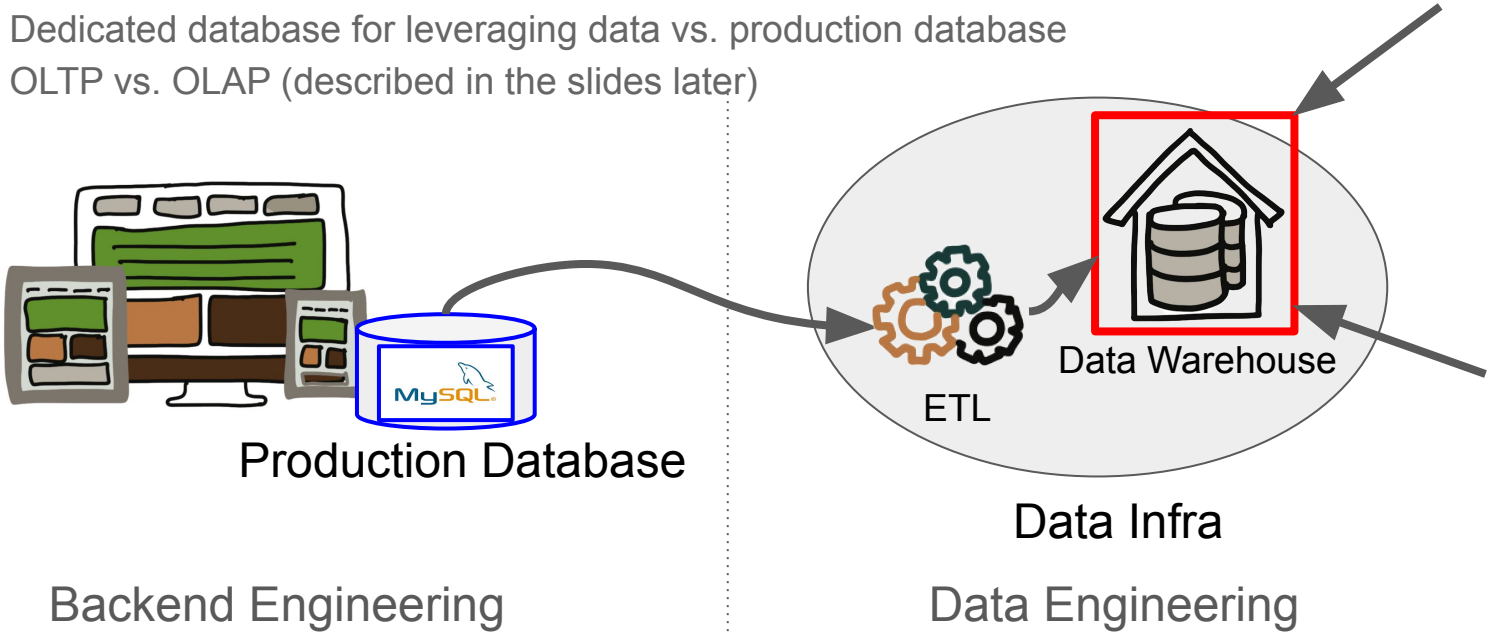
# Data Warehouse: Central Data Storage of your Company

- Central Database with all the data needed for the company
  - Stores all kinds of raw data from different sources
  - Still SQL based relational database (data modeling still matters)
- Mostly Cloud and Big Data based database
  - Size of data is more important than speed of data handling
  - Choose according to your situation (can be a MySQL at the beginning)



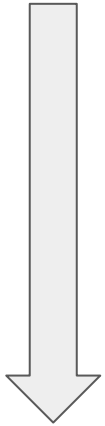
# Data Warehouse: A separate database only for Data Stuffs

- Building a data warehouse is the first step to becoming a true data org
- Data Warehouse should be separate from the production database.
  - Dedicated database for leveraging data vs. production database
  - OLTP vs. OLAP (described in the slides later)



# Data Infra: Foundation enabling data driven value generation

- Data Warehouse + Data Pipeline
  - The simplest form of Data Infra/Platform
- As the size of data increases
  - Data Lake is needed (more on this later)
  - Spark is needed as big data processing is required
- As data driven/informed decision matures
  - ELT solution is needed (“dbt”)
  - Dashboard solution is needed (Tableau, Superset, Power BI, …)
- As machine learning is being leveraged
  - ML related infra is required (NoSQL, API)
  - MLOps function is needed
  - Streaming data needs to be handled

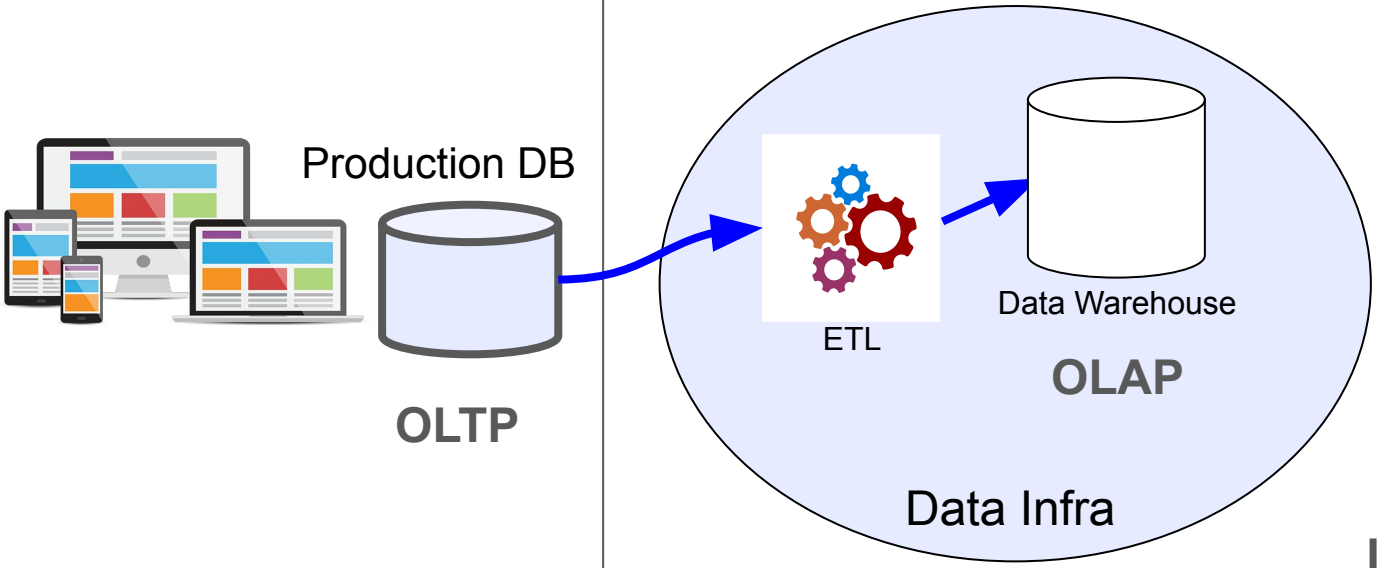


Sophistication of the infra

# OLTP vs. OLAP

# OLTP: OnLine Transaction Processing

# OLAP: OnLine Analytical Processing




**End Users, Customers**

Backend Engineers

Data Engineers

**Internal Audience**

# Difference between OLTP and OLAP

- OLTP: Production Database
  - Used to store data needed for operating the service/product
    - processing **speed** is more important
  - Managed by **Backend** engineering
- OLAP: Data Warehouse
  - Used to store data needed for data analysis and other data related work
    - **size of data** to handle is more critical
    - In other words, scalability is the key (and cost-effectiveness)
  - Managed by **Data** engineering

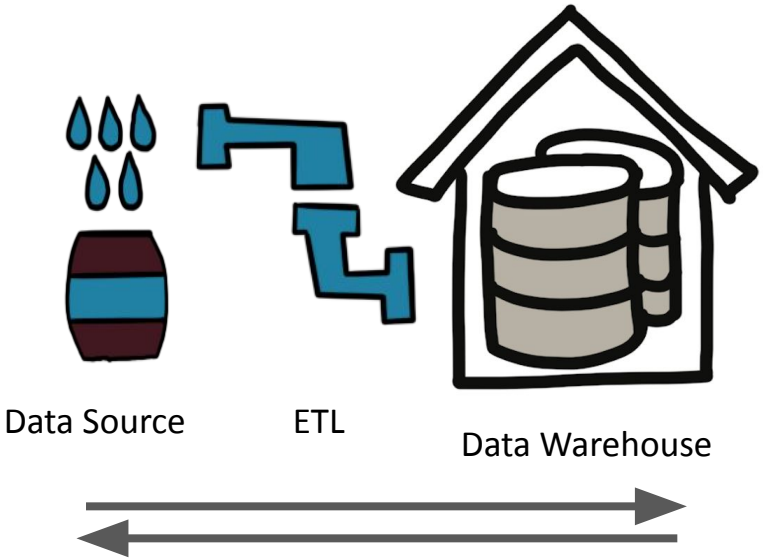
# What are OLTP Databases and OLAP Databases?

- OLTP
  - MySQL, PostgreSQL, Oracle, …
  - A single server solution with limited capacity
- OLAP
  - Snowflake, BigQuery, Redshift, Hive/Presto, ClickHouse, …
  - A group of servers behind the scene

# ETL vs. ELT

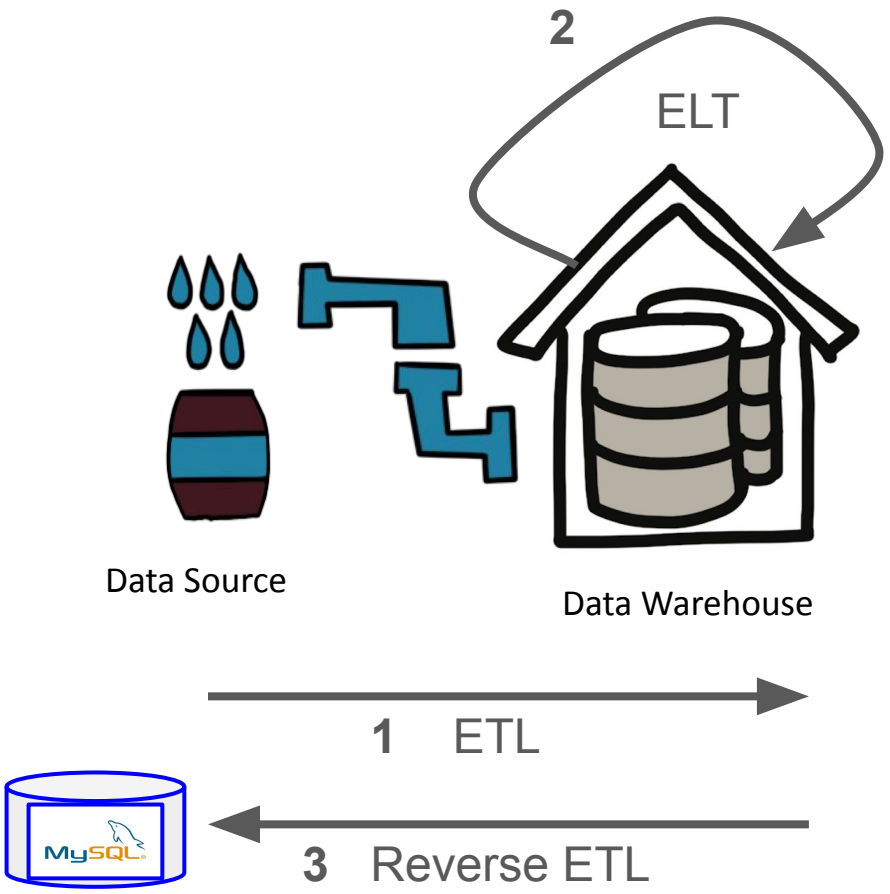
# Definition of Data Pipelines

- ETL: Extract, Transform and Load
- Transport data from sources to destination
  - Destinations: mostly Data Warehouse or Data Lakes
  - Done via coding or SaaS (FiveTran or StitchData)
- Data Source Examples:
  - Production databases, log files, APIs, …
- Data Destination Examples:
  - Data Warehouse, Data Lake, …
  - Production Systems (Reverse ETL)



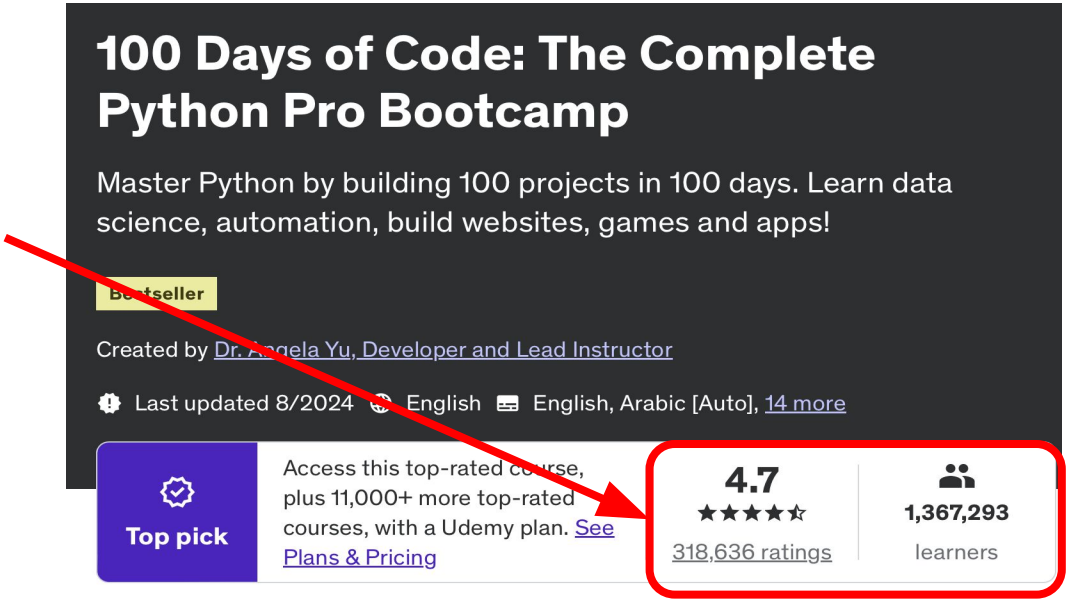
# Types of Data Pipeline

1. ETL
2. ELT (“T”)
3. Reverse ETL



# Reverse ETL example

1. Compute avg ratings, # of ratings and # of students per course in a data pipeline
2. Populate the results back to a production database, NoSQL and so on



# ETL vs. ELT

- ETL
  - The process of bringing data from outside the data system to inside
    - Creates Raw Data
  - Done by data engineers
- ELT
  - The process of manipulating data inside the data system to create new data (usually more abstracted and cleaned)
    - Creates cleaner and more usable abstract info
  - The most popular tool is dbt: “Analytics Engineer”
  - Done by data analysts
  - Input to ELT can be in the data lake (more on data lake in the later slides)

# Airflow Web UI Screenshot

Airflow | DAGs | Data Profiling | Browse | Admin | Docs | About

## DAGs

Search:

| | | DAG | Schedule | Owner | Recent Tasks |
|---|---|---|---|---|---|
| | On | Branch_to_Redshift | 15 11 * * * | john doe | 2 |
| | On | Build_Summary_Tables_Daily | 25 13 * * * | john doe | 6 |
| | On | Build_Summary_Tables_Hourly | 15,45 * * * * | john doe | 10 |
| | On | Cleanup_Logdir | 0 12,20,0 * * * | john doe | 2 |
| | On | Download_Gsheet | 30 14 * * * | john doe | 1 |
| | On | Export_Amplitude_To_Redshift | 30 12,20 * * * | john doe | 1 |
| | On | Export_Salesforce_To_Redshift | 30 11 * * * | john doe | 2 |
| | On | Postgres_to_Redshift_Daily | 30 12 * * * | john doe | 4 |
| | On | Postgres_to_Redshift_Hourly | 5,35 * * * * | john doe | 40 |
| | On | QuickBooks_to_Redshift | 25 12,0 * * * | john doe | 2 |

# Data Warehouse vs. Data Lake

# Weakness of Data Warehouse

- Not all data is structured (can’t be expressed as tables)
- What to do with unstructured data?
  - Unstructured data (ex: log files) tends to be much larger
  - Data Warehouse storage cost is too high to host unstructured data
- What if we can store unstructured data in a cheaper storage

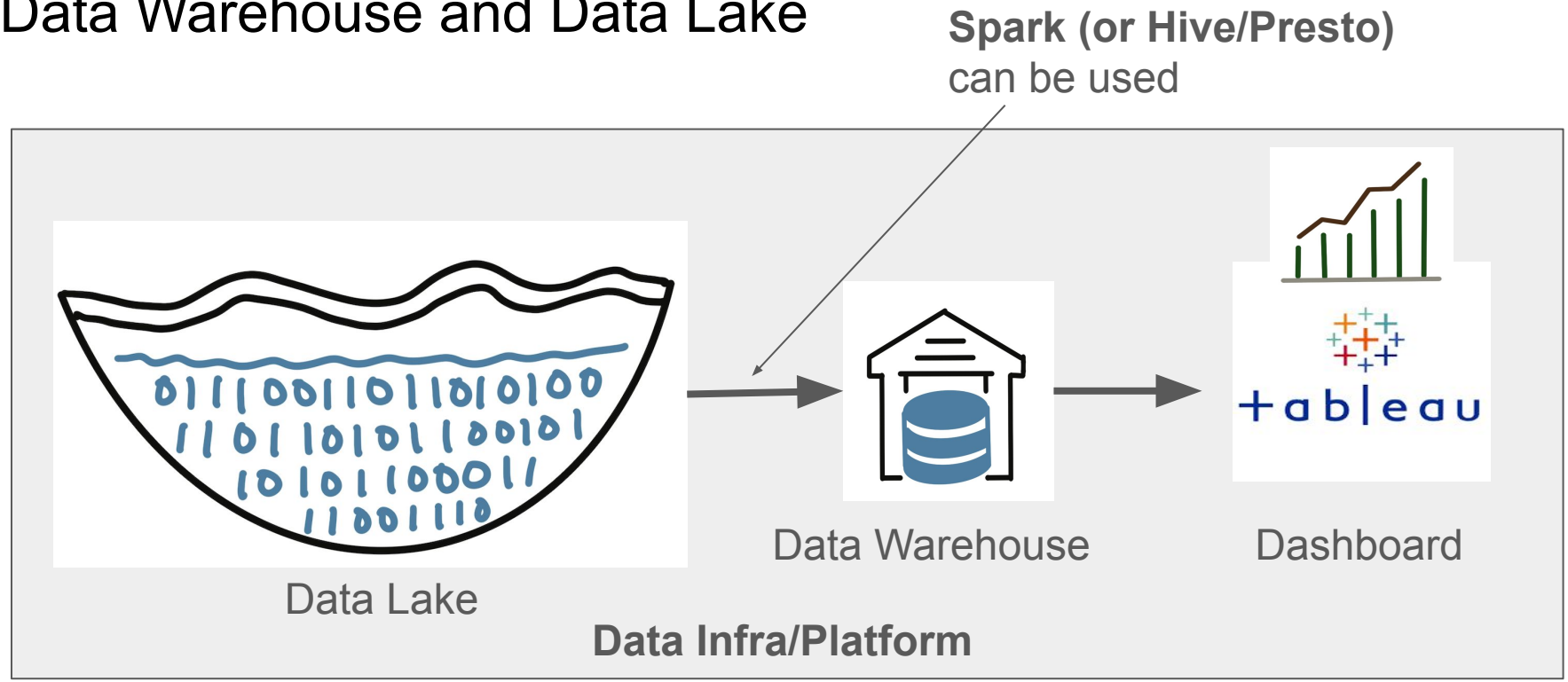
**Data Lake** to the rescue!

- Basically it is a very large file system (“Cloud Storage”)
  - In AWS, it would be S3 (Simple Storage Service)
  - In Google Cloud, it would be GCS (Google Cloud Storage)

# Data Warehouse vs. Data Lake

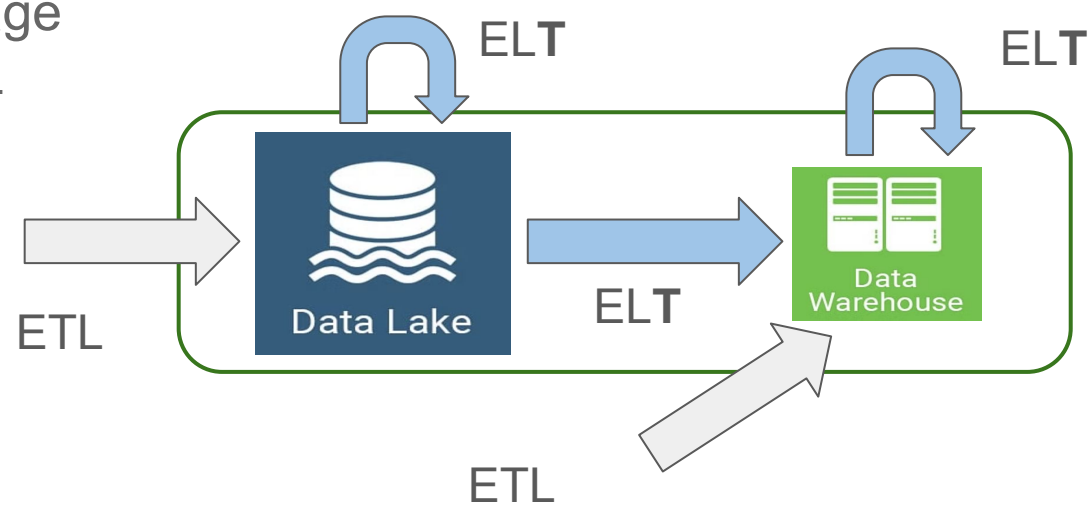
- Data Warehouse
  - Structured data is stored with some retention policy
  - Limited capacity with higher cost
- Data Lake
  - Structured data + Unstructured data
  - Order of magnitude bigger than DW
  - Cheap and big enough to store all the **raw** data without retention policy
- Both will likely be used together at some point
- LakeHouse
  - Data Lake + Data Warehouse
  - Iceberg from Netflix

# Data Warehouse and Data Lake



# Data Lake

- Order of magnitude bigger than Data Warehouse
  - Can store both **structured** data and **unstructured** data in cost effective fashion
  - Data can be stored in raw form without retention policy (as long as there are no legal issues)
- More like a cloud storage
  - AWS S3, GCP GCS, …



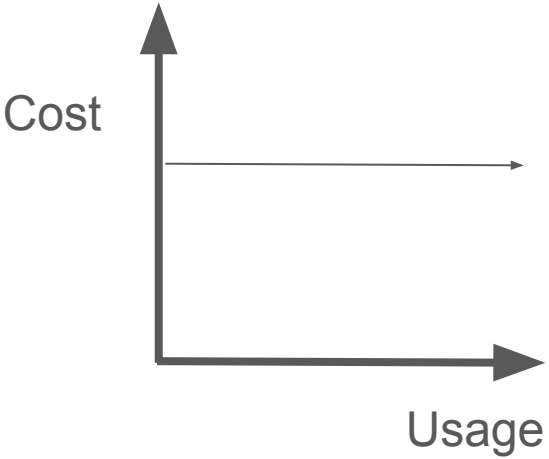
# Break

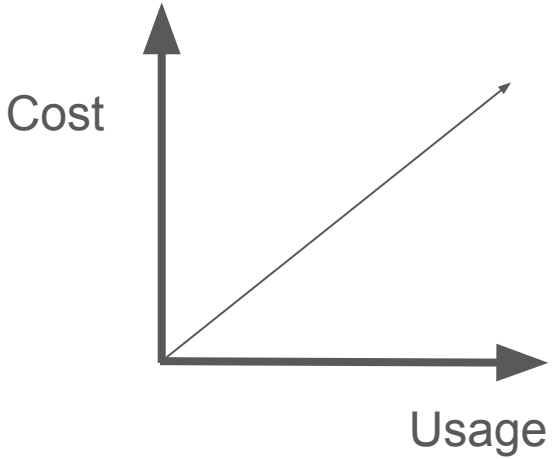
# Data Warehouse Options

# Data Warehouse: Different Options

- **Cloud** vs. On-Premise
- Open Source vs. Proprietary
  - Open Source ones are serviced from Cloud as well
- Fixed Cost vs. Variable Cost
  - CapEx (Capital expenditure) vs. OPEX (Operating expenses)

# Fixed Cost vs. Variable cost




- Pay a fixed amount monthly for a predetermined capacity
- Ex) AWS Redshift dc2.large (2 CPU, 15GB memory, 160GB SSD): $180/month

- Pay as you go (true Cloud fashion)
- More scalable option and easier to manage but the cost can go too high
- Separation of Storage and Compute

# What are the options

- AWS Redshift: Cloud, Proprietary, Fixed/Variable
- **Snowflake**: Cloud, Proprietary, Hybrid
- Google Cloud BigQuery: Cloud, Proprietary, Fixed/Variable
- Apache Hive: Cloud/On-Premise, Open Source, Fixed
  - Can work with Data Lake
- Apache Presto: Cloud/On-Premise, Open Source, Fixed
  - Can work with Data Lake
- Apache Iceberg: Cloud/On-Premise, Open Source, Fixed/Variable
  - This is a Data LakeHouse solution gaining momentum

Basically all of these are big data based Data Warehouse/Lake

# AWS Redshift

- AWS data warehouse started in 2012
  - Capable of distributed processing of PB-scale data
  - Supports PostgreSQL compatible SQL
  - Initially started with a fixed-cost model, but now also supports a variable-cost model (Redshift Serverless)
- Supports data formats such as CSV, JSON, Avro, Parquet, etc.
- Easy integration with other services within AWS
  - S3, DynamoDB, SageMaker, etc.
  - Supports execution of ML models (SageMaker)
  - Can be extended with services like Redshift Spectrum, AWS Athena and so on

# Google Cloud Bigquery

- Google Cloud's data warehouse service (launched in 2010)
  - A flagship service of Google Cloud
  - BigQuery SQL is powerful and complex
  - Supports both variable and fixed cost options
- Supports data formats such as CSV, JSON, Avro, Parquet, etc.
- Easy integration with other services within Google Cloud
  - Cloud Storage, Dataflow, AutoML, etc.

# Snowflake

- Started as a cloud-based data warehouse in 2011
  - Went public in 2020
  - Has now evolved to the point where it can be called a data cloud
  - Provides Data Sharing/Marketplace for selling data
  - Offers ETL and various data integration capabilities
- Enables big data storage, processing, and analysis based on SQL
  - Provides features for handling unstructured data and machine learning
  - Supports data formats such as CSV, JSON, Avro, Parquet, etc.
- Also supports S3, Google Cloud storage, and Azure Blob Storage

# Sneak peek at Snowflake

# Key Concept in Snowflake: Separation of Storage & Compute

- Snowflake separates storage from compute resources
- Compute resources are called "Virtual Warehouses"
- Virtual Warehouses can be scaled to different sizes based on computing power needs
  - We will be using X-Small by default

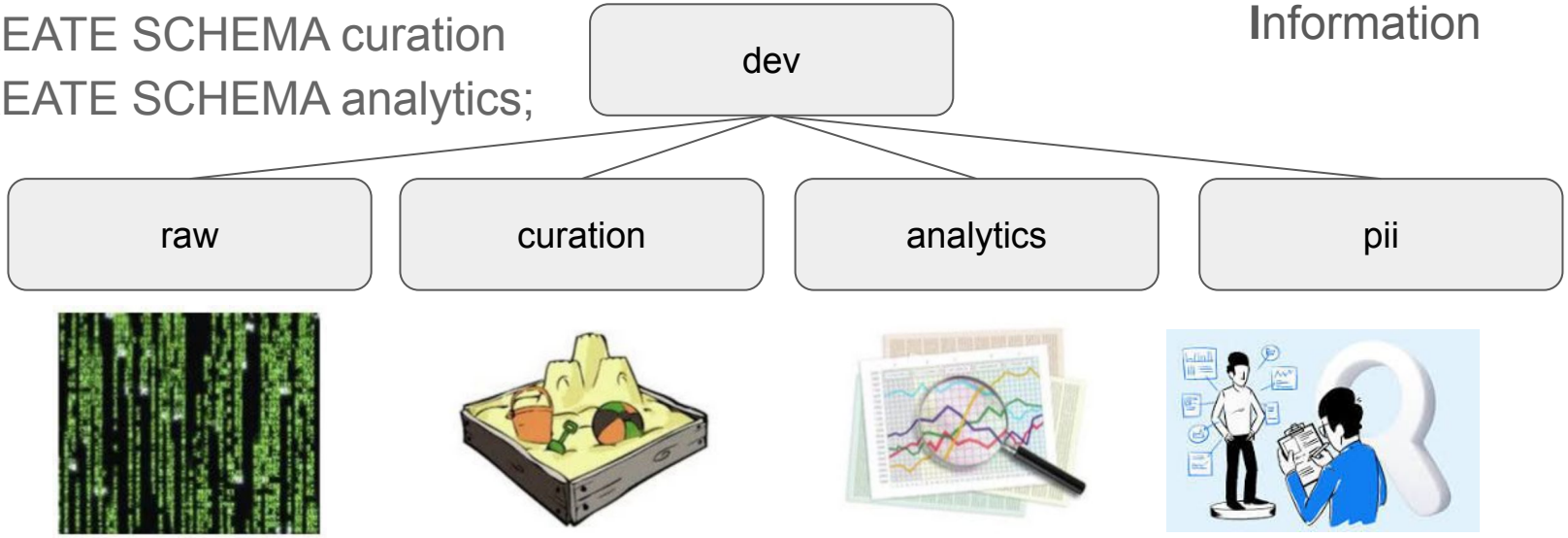
# Key Concept in Snowflake: Database & Schema Organization

- Each student will be assigned their own dedicated database
  - For example, USER_DB_FOX
- Within each database, we'll organize tables using multiple schemas:
  - Raw schema: Contains source data and ETL (Extract, Transform, Load) tables
  - Analytics schema: Contains processed data and ELT (Extract, Load, Transform) tables

# Database ➔ Schema: a way to organize your tables

CREATE DATABASE dev; (in our case USER_DB_FOX)
CREATE SCHEMA raw;
CREATE SCHEMA curation
CREATE SCHEMA analytics;

**P**ersonally
**I**dentifiable
**I**nformation




ETL Tables

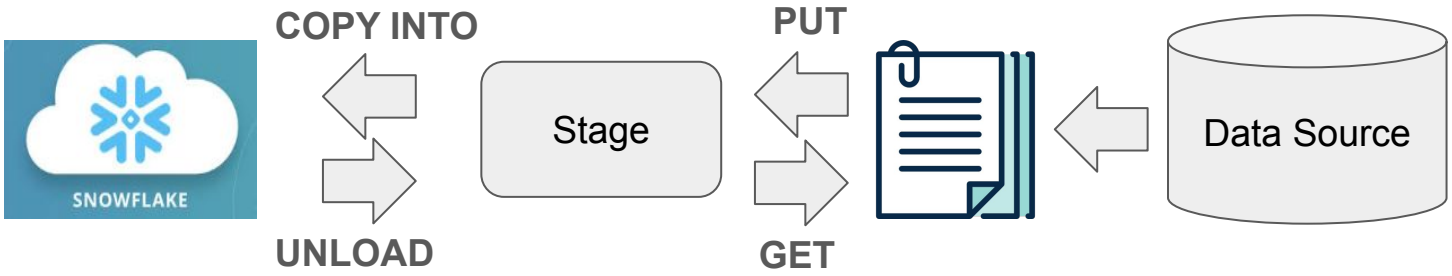
Cleanup

ELT Tables

PII Tables

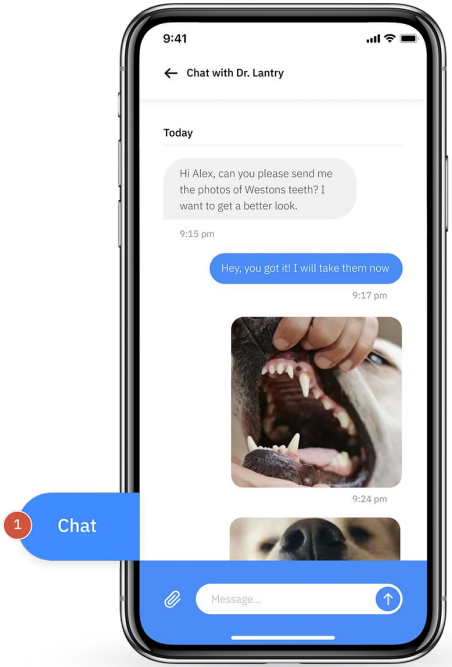
# Key Concept in Snowflake: Bulk Operations

- Handling Large-Scale Data Uploads
  - INSERT INTO becomes impractical when working with millions of records
  - Snowflake provides the COPY INTO command for efficient bulk data loading
  - This approach significantly improves performance compared to INSERT INTO
- File-Based Processing with Stages
  - Snowflake processes data at the file level rather than record-by-record
  - An intermediary storage location called a "stage" facilitates this process
  - Stages act as a temporary buffer between your data source and Snowflake tables



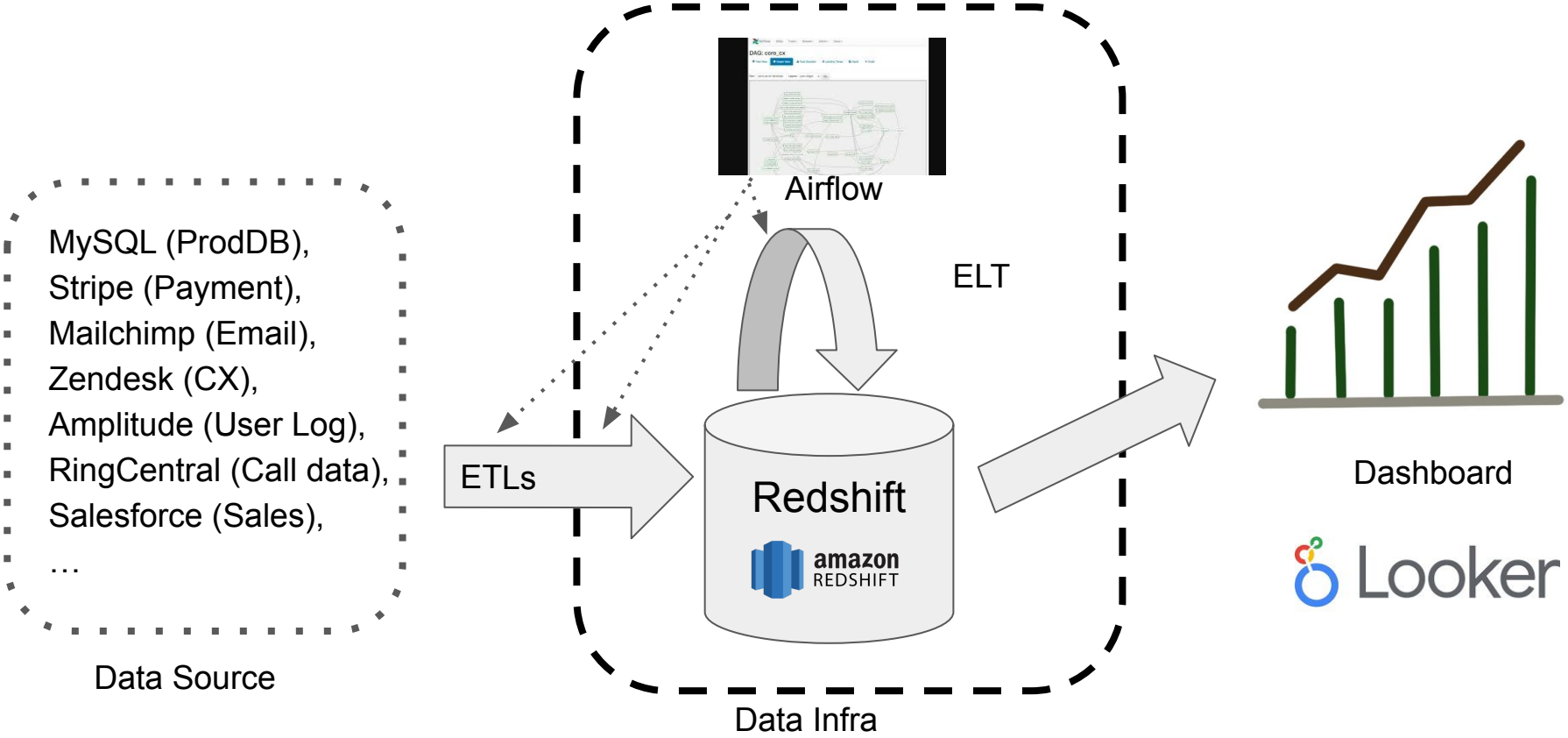
# Example Data Platform

# Case Study: Pet remote care marketplace service




- Matching Pet owners and Vets
- Components of its Data Infra
  - Airflow (Data Pipeline)
  - Redshift (Data Warehouse)
  - Looker (Dashboard)

# Data Infra


MySQL (ProdDB),
Stripe (Payment),
Mailchimp (Email),
Zendesk (CX),
Amplitude (User Log),
RingCentral (Call data),
Salesforce (Sales),
…
Data Source
ETLs
Airflow
ELT
Redshift
amazon
REDSHIFT
Data Infra
Dashboard
Looker

# Tables backed by ETL

## From MySQL

- Vet
- Hospital
- Vet_Hospital
- Client
- Client_Vet
- Pet
- Pet_Client
- Case
- Payment
- Address
- Video Calls
- Reviews

## From Amplitude

- Vet app
- Client app

## From Salesforce

- Account

## From Stripe (Credit Card Payment)

- Charge/PaymentIntent
- Refund

# Tables generated by ELT

- Payment Summary
- Case Summary
- Vet Summary
- Client Summary
- Hospital Summary
- Client Cohort Summary
- Vet Cohort Summary
- Hospital Cohort Summary

- These are used in the dashboards and in the reporting
- **dbt** is the most popular choice
  - vs SQL CTAS (Create Table … As SELECT)

# Lab 1

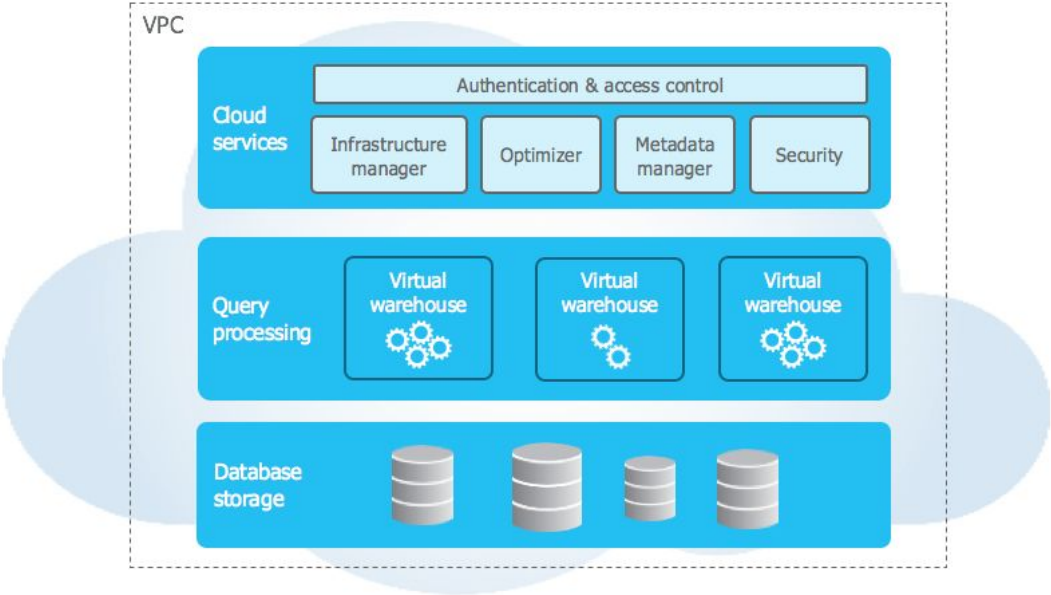
# Lab 1

- “ETL Pipeline + Forecasting Pipeline on Snowflake”
- Due on 3/05 (Week 7)
- Team Selection due by next week
  - https://sjsu.instructure.com/courses/1625908/groups#tab-114369
- More Details next Week

# Demo: Snowflake Setup & Basics

# Mini Recap on Snowflake

- Cloud-based data solution
- Separation of storage and compute
- SQL-based interface




https://docs.snowflake.com/en/user-guide/intro-key-concepts#snowflake-architecture

# Homework

# Homework #1

**Assignment - Snowflake Environment Setup**

Car Sales csv(synthetic) dataset is under the files/Homeworks/

1. (+1) Create Snowflake Account via Snowflake Academia as instructed
2. (+1) Create Schema 'schema_car_sales' under your Database. Use the warehouse of size XSMALL.
3. (+1) Create Stage 'stage_car_sales' in Snowflake and load the csv
4. (+2) Load Data into the 'data_car_sales' table (“COPY INTO” OR “UI LOAD”).
5. (+6) Write an SQL query to produce a report that shows the following insights
    1. The average number of cars sold per month in 2024, ordered Jan ➔ Dec
    2. The top 5 best-selling cars along with the count in 2024.
    3. The top 3 least-selling cars along with the count in 2024.

Submission Requirements:

- Submission should be a **PDF file.**
- Submission should comprise of Question, SQL Query and Snowflake Worksheet **Screenshot**.
- Snowflake Worksheet Screenshot should consist of query and results.

# Q & A
# Next Week: Snowflake (Deepdive)